

Fuzzy Logic-based Optimization of Flood Control System

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Abstract: Floods wreak havoc on the lives of people, making them a natural calamity of great significance. Flood catastrophes can have negative effects, including supplies and infrastructural destruction as well as fatalities. Urban floods can pose a major risk to public wellness, destroy property, and result in financial losses. To this level, it is imperative to improve flood control strategies through the installation of suitable facilities including detention reservoirs. Although smart systems may be used to regulate floods using detention reservoirs, there is a lack of research on the optimal design of such facilities in the face of flood threats. This paper presents a fuzzy logic-based control system for assessing the efficiency of flood drainage structures using three factors: hydrological efficiency, technological circumstance, and functional circumstance. The approach allows for a uniform evaluation of system dependability by synthesizing expertise information into 125 inference criteria. The system has been verified against experimental databases from drainage systems in Europe, and it performs well in a variety of functional and geographic scenarios. The platform has been coupled with Telegram and constructed with multiple detectors. Since the output needed for this research is in the type of values or linear equations, the Sugeno technique is more suited to be used in this investigation. Water stream and ultrasonic detectors are employed. The water stream detector determines the stream of water reaching the trial site at intervals of 0–11 liters per minute, while the ultrasonic detector measures the water concentration over a spectrum of 0–51 cm. With the help of the WiFi component and the NodeMCU ESP8266, statistics are transmitted to Telegram in real-time via the Firebase dataset. Water degree and release statistics are analyzed and Sugeno fuzzy logic is used for analyzing the outcomes. The mean deviation of the value from the ultrasonic detector, according to the investigation's findings, is 2.44%, or 97.59%. The mean inaccuracy displayed by the water stream detector is 0.207 liters per minute or an accuracy ratio of 87.07%.

Keywords: Flood, Urban flood, Disasters, Flood control system, and Fuzzy-logic.

I. INTRODUCTION

Cities are growing faster than ever before, which means that more and more precipitation needs to be evacuated from

them. Flooding in cities is also caused by the resistance of urban areas, the constrained ability of sewage systems, and the constrained capability of flooding collectors [1]. Metropolitan flood facilities network creation and management have an immediate effect on societal and ecological issues and are crucial to local government. The serious repercussions of urban floods, such as financial damages and limited funds for updating existing infrastructure, have made effective urban flooding construction imperative [2]. These issues are made worse by a changing climate, which modifies precipitation trends and, specifically, intensifies high flow. The ideal sewage size choice is a crucial component of floodwater system planning, which must strike a balance between hydraulic capability for rainfall transport and storage capacities to reduce the possibility of receiver overflow [3].

Subjection is frequently present in designers' perceptions of current or suggested designs, which causes uncertainty in decision-making. Because natural and artificial systems are innately unsure, imprecise, and ambiguous, fuzzy set concept and inference models (FIS) are used to transform expertise into sound decision rules [4].

Rainwater-runoff simulation, the initial stage of flooding layout, serves as the foundation for city development, water condition study, excess layout, and flood prediction. Due to the high processing costs of conventional models, scientists are looking into fuzzy logic-based techniques for performance and simplicity. For example, drainage estimations employing past data have been enhanced by the combination of principal component assessment (PCA) adaptable neuro-FIS (ANFIS) [5]. ANFIS's adaptation is furthermore illustrated in electrical networks, where it combines the flexibility of fuzzy logic with the effectiveness of neural network development to surpass traditional techniques in optimal energy peak surveillance for solar panels [6].

Excessive precipitation, poor ground permeation, and antiquated systems all contribute to urban floods. To avoid

channel excess, efficient floodwater draining structures must take into account growing catchments, shifting land utilization, and contaminant levels. Urban design has also progressed with the use of fuzzy logic and GIS [7]. Others showed ways fuzzy-GIS systems handle inaccurate geographical information for ecological condition assessments, while others classified socioeconomic and ecological factors into uniform areas employing fuzzy rule-based categorization. These methods demonstrate how flexible fuzzy logic is in managing hydrological and geographical uncertainty [8].

The Fuzzy Mamdani technique is a component of the FIS that uses the MAX-MIN functioning to determine the optimum course of action in an ambiguous issue. The system results in an integer rather than a fuzzy collection. Since the output needed for this research is in the type of values or linear formulas, the Sugeno technique is thus better suited to be used in this investigation [9].

The free program for sending texts is called Telegram. Telegram is compatible with Linux, Windows, and Mac OS and may be used on handheld gadgets. The chatbot characteristic is among the APIs offered. The use of chatbots has begun to spread significantly. One benefit of chatbots is their capacity to reliably provide consumers with data that is not time-limited. Additionally, anybody can build a bot that will respond to each of its clients if they provide an instruction text that the chatbot can understand. Telegram encoding has been implemented, making statistics transmission secure [10].

In light of this context, this study used Sugeno's fuzzy logic for modeling the river testing setting and creating a river surveillance instrument for flood control that predicts the water threshold per unit of duration. So that individuals are prepared to control flood disasters promptly, this instrument may swiftly and efficiently convey flood alert data to the general population using the Telegram Chatbot medium.

II. LITERATURE REVIEW

The goal of the study by [11] is to track the climate in real-time. The detectors for humidity, airflow, and moisture are used in the structure's architecture. In order to reduce declines from floods, a study by [12] attempts to track or control stream water flows. Through Android and a web page, the structure is intended to show data on stream and rainwater level management in specific locations. The goal of the study by [13] is to assist householders in identifying floods early on so they may relocate or preserve vulnerable elements that are susceptible to water in a more secure spot.

The technique is intended to provide people with warning messages about flooding through SMS. The goal of this study by is to provide an online database with statistics on flood conditions and water threshold statistics that is secure, warning, and harmful.

An Investigation into a Microstrip Patch Antenna operating on the N77 Band Through the Utilization of Slot Loading Based on Simulation [14-15].

Nearly 25% of US air passenger travels in 2019 were foreign, completely on US airlines. This represented 4.5% of US person-miles. US travel demand surveys and models rarely include overseas travel. How will autonomous vehicles change long-distance travel in Texas [16-17]

Methods for producing energy from solar panels by utilizing statistical analysis, a number of cutting-edge tools and procedures, including the IFS Weighted Averaging Operator. Several control mechanisms are employed in a multi-area power system to regulate the load frequency. For the purpose of charging electric vehicles, a fuzzy logic controller is employed to offer implicit control of a DC motor via a DC-DC converter. The Load Frequency Control Scheme takes advantage of both PID and MPC [18-23]. Fuzzy deep learning AI is being developed for real-time decision-making and edge computing in uncertain Internet of Things conditions. Renewable Energy Applications of Ambiguous Fuzzy Hybrid Averaging Operators. This study reviews green energy and intuitionistic fuzzy decision-making studies [24-26].

For the purpose of removing speckles from echocardiographic images, a hybrid fuzzy filter was applied. The book titled "Deep Learning: Current Trends and Techniques" was included in the publication that was released in Boca Raton and was titled "Deep Learning in Visual Computing and Signal Processing" [27-29]. The detection of digital deception is being accomplished by the utilization of a method for deepfake detection that is a combination of CNN and RNN. In this context, an evaluation of transfer learning for visual multiple target tracking is being accomplished. For the purpose of achieving better performance, the process of emotion recognition was carried out with the assistance of a hybrid model that included CNN-LSTM and Transformer. The Quaternion Intuitionistic Fuzzy Fusion Process is where the discussion of applications to the classification of photovoltaic solar power plants [30-33] takes place.

The possibilities of fuzzy logic are further demonstrated by recent developments in real-time management: improved flood prevention in sewer systems utilizing fuzzy rules refined by genetic algorithms, resulting in a 25 percent decrease in flood levels. Additionally, improved peak flow regulation by up to 87 percent in the presence of unclear precipitation estimates by developing a forecasting fuzzy-rule framework for flooding preservation. Nevertheless, without combining hydraulic, architectural, and functional elements into a coherent structure, these investigations mostly focus on particular facets of flood administration, like reducing flooding or excess control.

III. MATERIALS AND METHODS

A. Fuzzy logic and Fuzzy Set

Dr. Lotfi Zadeh developed fuzzy logic in the 1960s as an analytical structure to deal with the inaccuracy and ambiguity present throughout numerous real-world structures. Fuzzy logic expands on the stringent evaluation of statements as true (1) or false (0) found in traditional Boolean logic by permitting truth levels to continually range between 0 and 1. This feature captures the subtleties of ambiguity seen in natural processes by allowing assertions to be both partly true and partly incorrect.

By using language concepts and associated association operations, an FIS uses fuzzy set concepts to translate input to output choices. The FIS creates a strong framework for establishing decisions in unpredictable situations by incorporating expertise using a set of fuzzy guidelines. The Mamdani FIS is used in this work because of its easy-to-

understand rule-based architecture and track record of success in technical uses.

The conventional idea of set inclusion is expanded upon in the context of the fuzzy collection concept. A fuzzy collection A can be described as an array of organized couples provided a universal list U , frequently recognized as a potential space:

$$A = \{(x, u_A(x)) | x \in X\} \quad (1)$$

The description of a crispy (or sharpening) collection in which association is binary, stands in stark comparison to this. The membership operation for a crisp collection is provided by:

$$X_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A, \end{cases} \quad (2)$$

Therefore, fuzzy sets permit partial membership, whereas conventional collections are distinguished by distinct, exclusive borders. This characteristic offers a versatile and subtle system for simulating the ambiguous borders that are frequently present in real-world uses.

B. Linguistic Factors

An FIS uses linguistic parameters to use membership processes to convert input numbers into subjective groups. These operations specify how closely an empirical number fits within a particular linguistic group. Three input factors (hydraulic efficiency, technological circumstance, and operating circumstance) and one output factor (result) are taken into account in the fuzzy logic framework that was constructed.

C. Fuzzy Platform Rule

A collection of inference guidelines intended to assess the floodwater draining method's functional state is used to establish the fuzzy regulator. To identify the proper outcome group, these rules combine the three input parameters. An entire set of 125 criteria were developed in light of the number of conceivable input pairs.

Particularly critical action is necessary due to significantly deteriorated hydrologic efficiency and poor technological and functional circumstances (Quick Substitution Needed). Based on how the three input factors communicate, a modest system state usually translates into either regular service or surveillance alone. No Intervention Needed refers to a structure that is in outstanding form and has functional variables that are performing properly.

The application phase of the inference function involves creating rules for processing fuzzy input. The system has two inputs: water flow and release, with 5 and 3 parameters respectively, totaling 15 factors. The fundamental laws of fuzzy logic are as follows: 1) Water location with secure threshold and minimal stream extent; 2) Water location with secure threshold and moderate stream pace; 3) Water location with secure degree and substantial stream pace; 4) Water location with waiting threshold and minimal stream pace; 5) Water location with waiting degree and moderate flows; 6) Water location with a high stream pace and waiting degree; 7) The location of water with a minimal degree of threat and an average stream pace; 8) The location of water with a

medium water stream pace and an appropriate degree of threat; 9) Water location with high water stream pace and threat degree; 10) Water location with low stream percentage and warning degree; 11) Water location with moderate water stream threshold and warning threshold; 12) Water location with high water stream threshold and warning degree; 13) Water location with minimal water stream pace and flood degree; 14) Water location with moderate water stream level and flood degree; 15) Water location with high water stream percentage and flood depth.

The result of every rule is produced during the aggregation phase. The values of the five fuzzy logic results range from 0 to 1. The flood estimation framework's four outputs Secure Data, Warning, Threat, and Flood have measurements: 1) Secure Data is equal to zero; 2) Warning Data is equal to 0.26; 3) The threat Data is equal to 0.6; 4) Warning Data is equal to 0.76; and 5) Flood Data is equal to one.

When scaled by the water release factor, every water level factor produces two distinct outputs, with the exception of the flood factor, which only produces one result. The current state of one water threshold factor can generate the subsequent water threshold factor for all of them, yielding 27 fuzzy value regulations from 2 sources with 5 water degree factors and 3 water release factors.

D. Water Level Rise Estimation Model

The duration at which the water flows intensely affects the plan of the estimation of the rise in water degree. Therefore, to compute the estimation of the rise in water employing the equation for the duration of water extent, use the form of the equation for water release as subsequent.

$$W = \frac{v}{D} \quad (3)$$

E. Real-time Database for Validation

A database from actual research on flood drains was used to verify the suggested fuzzy logic framework. These databases offer alternative viewpoints on system efficiency and breakdown processes since they were gathered from various functional and geographical settings. Although the databases were combined into a single model for uniformity, their different histories guarantee a strong assessment of the system's generalization.

The database which focuses on flood drainage systems in European metropolitan contexts, comes from a technological review. This dataset includes thorough examinations of pipeline sections that record architectural flaws like joint shifts, lengthwise cracks, root incursions, and leaking trends. With reported flaws closely correlated with the necessity of service, the database highlights architectural deterioration and its effect on hydrologic performance.

IV. RESULT AND DISCUSSION

The NodeMCU 8266 microprocessor serves as both a controlling device and a Wi-Fi connection source for the purpose of receiving water stream detectors and ultrasonic detector values. To measure the water threshold, an ultrasonic detector is positioned in the middle of the lid, which has an opening in it. The ultrasonic detector, which has an opening for collecting water and receives water stream statistics via a

valve linked from the water resource to the water stream detector, is positioned adjacent to the water stream detector. Following the reading of the outflow and water degree statistics, the microprocessor will transmit a real-time dataset to Firebase, show the readings on the 16*2 I2C LCD, and transmit the Telegram Automation alerts that are autonomously generated by the instrument.

A test vessel containing imitation vegetation and river boulders is used for river modeling to replicate the appearance of the real river. In addition to holding water, the test vessel serves as a benchmark for determining the volume of water that will be employed when the flood detection instrument is operational.

Sugeno's fuzzy logic modeling flood estimation framework in this research makes utilization of MATLAB R2015a instruments, which include one output (water flow data) and two inputs (water flow and outflow).

A. Deployment of Water Flow Rise Estimation

In this research, forecasts for rising water volumes are applied using Arduino IDE coding and a water quantity temporal equation that was adjusted to meet the requirements of the apparatus.

A rise in water levels from secure to a waiting threshold, threat to a warning threshold, or flood threshold is predicted using a flood estimation computation. The secure-to-waiting socks are 21 cm, the waiting-to-threat is 11are cm, the threat-to-warning is 11 cm, the warning-to-flood is 6 cm, and the flood has a 6 cm period from the highest point of the experimental vessel. This computation applies to every threshold from 5 estimation uses at 6 water scales.

By evaluating 600 distinct flooding situations, the suggested stochastic optimization approach was used to identify the best geometric characteristics for the reservoirs with various kinds of outlets. The proposed stochastic optimization design's Pareto-optimal approaches for (a) rectangular weirs, (b) orifices, (c) proportional, and (d) triangular are shown in Fig.1.

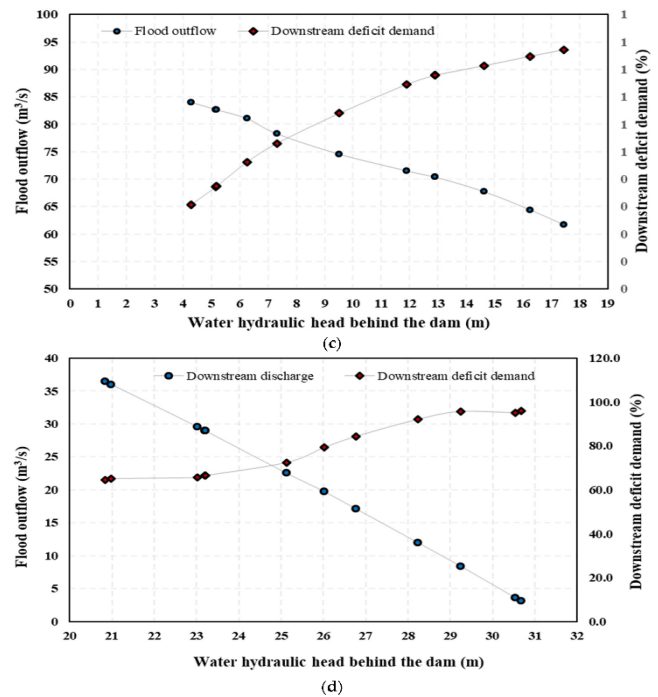
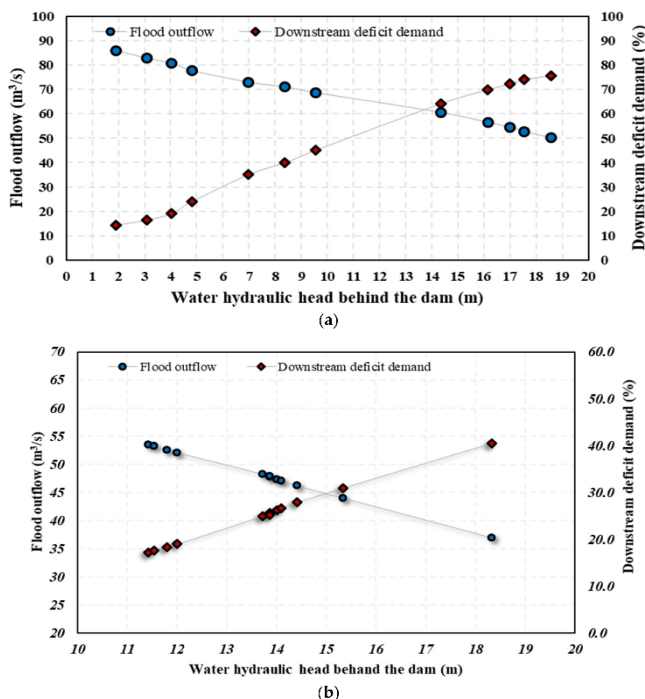


Fig.1. The proposed stochastic optimization design's Pareto-optimal approaches for (a) rectangular weirs (b) orifices, (c) proportional, and (d) triangular.

B. Ultrasonic Detector Accuracy Assesment

Table I shows the results of validating the ultrasonic detector's accuracy in comparison to a measurement ruler. The outcomes of the assessment between the measurement ruler and the ultrasonic device are as follows.

TABLE I ULTRASONIC DETECTOR ACCURACY ASSESMENT

No	Detector Score		Error Variance		Accuracy
	Ruler $\pm$ (cm)	Ultrasonic (cm)	(%)	$\pm$ cm	
1	18.53	20	2.6	0.49	97.51
2	21.3	23	3.8	0.9	96.31
3	23.93	25	0.04	0.09	99.98
4	31	32	3.34	1.1	96.68
5	26.8	27	2.7	0.8	97.41

The performance of the ultrasonic detector is tested in Table I employing five experimental specimens to measure the measurement significance of the instrument statistics with a ruler. This yields a mean error variance of 2.44% and an instrument accuracy of 97.59%.

Two test elements are used to evaluate the water stream detector: determining the water stream reaching the experiment vessel and assessing the detector's performance.

C. Verifying the Water Stream Reaching the Experimental Vessel

Employing a water stream detector, the water stream is tested. Markers of accomplishment are used to retrieve statistics, specifically the ability to comprehend water that enters via a hose that is linked to the detector when water is streaming. Employing a hose connected from the water supply to the water stream detector's switch, the flood estimation instrument is positioned at the highest point of the experimental vessel by creating a hole for the water stream detector. This allows the detector to direct water from the supply to the trial vessel. The water stream detector is used by the switch to determine the arriving water stream.

Next, using water that is stored in the water storage (tower), the experiment is conducted by turning on the water faucet. After that, the water supply opens for 30 secs using a 45-degree lever gauge. Table II displays the outcomes of the water stream detector experiment.

TABLE II WATRE STREAM DETECTOR ASSESSMENTS

No	Analysis Data	Water Stream Detector Data (L/min)	Time (secs)
1	Analysis 1	3.24 3.14 3.28 3.36	30
2	Analysis 2	2.40 1.91 1.81 1.83 1.90	30
3	Analysis 3	2.0 2.13 1.84 1.74 1.44	30
4	Analysis 4	0.74 1.02 0.88 0.82 0.80	30
5	Analysis 5	1.2 1.04 1.21 1.34	30

Five experimental specimens with varying water stream detector readings are taken from Table II and have the water tap gauge adjusted at 45° for 30 sec.

D. Analysis of Water Stream Detector Performance

Table III shows the variation when analyzing water release statistics from a water supply (a water tap) with a water tap's gauge that opens at 45° for 30 seconds to evaluate the performance of the water stream detector.

TABLE II PERFORMANCE OF WATER STREAM DETECTOR ASSESSMENTS

Analysis Data	Water Stream Detector Data (L/min)	Error Variance		Time (secs)	Accuracy (%)
		L/min	(%)		
Analysis 1	3.24	0.2	3.3	30	96.81
	3.14	0.25	4.30		95.72
	3.28	0.09	2.40		97.63
	3.36				
Mean Value		0.2	3.30		96.72
Analysis 2	2.40	1.50	78.39	30	21.63
	1.91	0.2	5.59		94.43
	1.81				
	1.83	0.06	2.66		97.36
Mean value		0.43	22.20		77.82
Analysis 3	2.0			30	
	2.13	0.4	16.40		83.62
	1.84	0.10	5.18		94.84
	1.74	0.4	20.84		79.18
Mean value		0.22	12.37		87.65
Analysis 4	0.74	0.32	30.40	30	69.62
	1.02	0.15	15.92		84.10

	0.88 0.82 0.80	0.09 0.17	11 16.14		91 83.88
Mean value		0.15	16		86
Analysis 5	1.2	0.09	7.85	30	92.17
	1.04	0.18	14.30		85.72
	1.21				
	1.34	0.12	8.88		91.14
Mean Value		0.17	12.56		87.86

Table III examines the performance of the water-stream detector using five examination specimens. The measurement of the instrument's statistics is tested using a water faucet gauge that tilts at 45° for 30 secs, resulting in a mean examination error variance of 0.207 L/min and a performance proportion of 87.03%. Table IV displays the findings of the total contrast of the input and output fuzzy data on the instrument after evaluation with MATLAB.

TABLE IV EVALUATION OF ASSESSMENTS ON INSTRUMENTS WITH FUZZY SCORES USING MATLAB

Input Data		Output Data Fuzzy			Details
Water depth	Flow level	Score	Water Depth	Warning Water Depth	
36	4.19	0.626	Threat	Threat	Correspondence
38	4.16	0.626	Threat	Threat	Correspondence
41	4.01	0.626	Threat	Threat	Correspondence
22	4.32	0.176	Waiting	Waiting	Correspondence
27	3.82	0.376	Waiting	Waiting	Correspondence

Table IV provides five data points from which the researchers deduced that their analysis of readings on the instrument with fuzzy outcomes on MATLAB yielded adequate findings since the fuzzy levels with water data were comparable to the water data on the instrument.

V. CONCLUSION

In recent years, urban flooding fatalities have increased due to shifting climates. Flood facility management decision-support is enhanced by the fuzzy-inference framework. By combining Hydraulic Efficiency, Technological Circumstance, and Functional State, the simulation manages real-world variability and complexity. Due to its accuracy, tolerance for variable changes, and ability to manage several databases, utility executives and engineers find it useful. Adding the fuzzy logic controller to their hydrological simulation program can help local governments find pipelines that require extra inspection or maintenance. The consequence is less urban floods, improved service plans, and more efficient use of scarce cash. Openness and rule-based structure improve participant trust and ensure proposals are understood and implementable. This study's results are: Five specimens are used to measure the ultrasonic detector's statistics using a ruler, resulting in a mean error variation of 2.44% and instrument accuracy of 97.59%. The water stream detector's accuracy is evaluated using 5 specimens and a water tap handle paved at 45° for 30 seconds. The mean variation in the measurement error is 0.207 L/min and the proportion of instrument accuracy is 87.07%. With fuzzy data with water data similar to the instrument's water data, MATLAB's instrument reading evaluation results are suitable for Sugeno's Fuzzy Logic assessment. Future study should

address data confidentiality, expertise biases, and fair access in system implementation. By ensuring the controller performs regionally, protecting private information, and decreasing expert-based rule bias, the fuzzy logic paradigm can be a just and efficient instrument for urban facility management. As a decision-support platform rather than a final decision-maker, it highlights the need to combine technical outcomes with broader societal and ecological considerations for fair and enduring results.

REFERENCES

- [1] Qi W, Ma C, Xu H, Chen Z, Zhao K, Han H. A review of applications of urban flood models in flood mitigation strategies. *Natural Hazards*. 2021 Aug;108:31-62.
- [2] Luo P, Luo M, Li F, Qi X, Huo A, Wang Z, He B, Takara K, Nover D, Wang Y. Urban flood numerical simulation: Research, methods, and future perspectives. *Environmental Modelling & Software*. 2022 Oct 1;156:105478.
- [3] Agonafir C, Lakhankar T, Khanbilvardi R, Krakauer N, Radell D, Devineni N. A review of recent advances in urban flood research. *Water Security*. 2023 Aug 1;19:100141.
- [4] Ullah I, Kovacs G, Lenner T. Impact of climate change on flood risk with urban resilience perspective using fuzzy logic model. *Earth Science Informatics*. 2025 Jun;18(2):346.
- [5] Shaikh MP, Yadav SM, Manekar VL. Flood hazard mapping by linking CF, AHP, and fuzzy logic techniques in urban areas. *Natural Hazards Review*. 2024 Feb 1;25(1):04023048.
- [6] Sun L, Xia J, She D, Ding W, Jiang J, Liu B, Zhao F. A predictive fuzzy logic and rule-based control approach for practical real-time operation of urban stormwater storage system. *Water Research*. 2024 Nov 15;266:122437.
- [7] Baalousha HM, Younes A, Yassin MA, Fahs M. Comparison of the fuzzy analytic hierarchy process (F-AHP) and fuzzy logic for flood exposure risk assessment in arid regions. *Hydrology*. 2023 Jul;10(7):136.
- [8] Bouamrane A, Derdous O, Dahri N, Tachi SE, Boutebba K, Bouziane MT. A comparison of the analytical hierarchy process and the fuzzy logic approach for flood susceptibility mapping in a semi-arid ungauged basin (Biskra basin: Algeria). *International Journal of River Basin Management*. 2022 Apr 3;20(2):203-13.
- [9] Naikoo MW, Talukdar S, Das T, Rahman A. Identification of homogenous rainfall regions with trend analysis using fuzzy logic and clustering approach coupled with advanced trend analysis techniques in Mumbai city. *Urban Climate*. 2022 Dec 1;46:101306.
- [10] Boota MW, Zwain HM, Shi X, Guo J, Li Y, Tayyab M, Soomro MH, Hu C, Liu C, Wang Y, Wahid JA. How effective is Twitter (X) social media data for urban flood management? *Journal of Hydrology*. 2024 May 1;634:131129.
- [11] Chaduvula K, Markapudi BR, Jyothi CR. Design and Implementation of IoT-based flood alert monitoring system using microcontroller 8051. *Materials Today: Proceedings*. 2023 Jan 1;80:2840-4.
- [12] Ulum MB, Badri F. Sistem Monitoring Cuaca Dan Peringatan Banjir Berbasis Iot Dengan Menggunakan Aplikasi Mit App Inventor. *JITET (Jurnal Informatika dan Teknik Elektro Terapan)*. 2023;11(3):319-28.
- [13] Hasiri EM, Allia HN. Peringatan Dini Banjir Menggunakan Multi Sensor Pada Prototype Aliran Sungai Berbasis Internet of Things. *Jurnal Informatika*. 2023 Jun 21;12(1):60-9.
- [14] S. Ara and P. K. Nunna, "Ultrawide band High Gain Inset Feed Simple Structure in Patch Antenna for Sub 6-GHz Application," in *Proc. 2024 DICCT*, Dehradun, India, 2024, pp. 698-702, doi: 10.1109/DICCT 61038.2024.10532981.
- [15] S. Ara, et al., "Simulation Based Analysis of Microstrip Patch Antenna for N77 Band Using Slot Loading," in *Proc. DICCT*, Dehradun, India, 2024, pp. 629-632, doi: 10.1109/DICCT61038.2024.10532853.
- [16] P. Paithankar, et al., "International travel patterns: exploring destination preferences and airfare trends to and from the USA," *Transportation Planning and Technology*, vol. 47, no. 8, pp. 1243-1261, 2024. DOI: 10.1080/03081060.2024.2366241.
- [17] P. Paithankar, et al., "Impact of autonomous vehicles on long-distance travel mode and destination choices in Texas," *Research in Transportation Economics*, vol. 110, p. 101521, 2025, doi: 10.1016/j.retrec.2025.101521.
- [18] A. Dagar, et al., "Electrical Power Plant Decision-Making by Utilizing IFS Weighted Averaging Operator," in *Proc. IITCEE*, Bangalore, India, 2024, pp. 1-4, doi: 10.1109/IITCEE59897.2024.10467655.
- [19] S. Dhull, et al., "Intelligent Tools and Techniques for Photovoltaic Solar Energy Generation Plants with Statistical Analysis," in *Proc. 2024 I2CT*, Pune, India, 2024, pp. 1-6, doi: 10.1109/I2CT61223. 2024.10543971.
- [20] I. Ahamad et al., "Load Frequency Control in Multi-Area Power System Using Different Control Schemes," in *Proc. 2024 IC3TES*, India, 2024, pp. 1-6, doi: 10.1109/IC3TES62412.2024.10877655.
- [21] S. Semwal, et al., "Implicit Control of DC-DC Converter based DC Motor for the Charging of Electric Vehicle using Fuzzy Logic Controller," in *Proc. GCAT*, Bangalore, India, 2024, pp. 1-6, doi: 10.1109/GCAT62922.2024.10923979.
- [22] A. Singh et al., "Model Predictive Control (MPC) and Proportional-Integral-Derivative (PID) Controllers for Load Frequency Control Scheme," *Suranaree J. Sci. Technol.*, vol. 32, no. 1, 2025.
- [23] S. Semwal, et al., "Comparative Analysis of Two-Area Load Frequency Control Using Tilted-Integral-Derivative (TID) and Proportional-Integral-Derivative (PID) Controllers," in *Proc. ICTEST*, Kochi, India, 2024, pp. 1-5, doi: 10.1109/ICTEST60614.2024. 10576133.
- [24] B. P. Joshi et al., "Fuzzy-Deep Learning-Based Artificial Intelligence for Edge Computing and Real-Time Decision-Making in Uncertain IoT Environments," in *Proc. 2025 CE2CT*, India, 2025, pp. 1301-1306, doi: 10.1109/CE2CT64011.2025.10941321.
- [25] N. Singh, et al., "Renewable-Energy-System Applications of Ambiguous-Fuzzy-Hybrid-Averaging Operator," in *Proc. ICICACS*, Raichur, India, 2024, pp. 1-6, doi: 10.1109/ICICACS60521.2024. 10498184.
- [26] B. P. Joshi, et al., "A systematic literature review on Intuitionistic Fuzzy Decision Making with Applications in Green Energy," *2024 ICACCM*, 2024, pp. 1-5, doi: 10.1109/ICACCM61117.2024. 11059111.
- [27] A. Balodi, "Despeckling in echocardiographic images using a hybrid fuzzy filter", In *Image Processing for Automated Diagnosis of Cardiac Diseases*; Elsevier: Amsterdam, The Netherlands, 2021; pp. 77-97.
- [28] A. Balodi, et al., "Computer-Aided Classification of the Mitral Regurgitation Using Multiresolution Local Binary Pattern", *Neural Comput. Appl.* 2020, 32, 2205-2215.
- [29] B. Sharma, et al., "Deep Learning: Current Trends and Techniques," in *Deep Learning in Visual Computing and Signal Processing*, Boca Raton : Apple Academic Press, 2022, pp. 55-69. doi: 10.1201/9781003277224-3.
- [30] C. Sharma et al., "An Efficient Algorithm for Generating Interactive Visualization of Large Datasets," in *Proc. 2025 CE2CT*, India, 2025, pp. 441-445, doi: 10.1109/CE2CT64011. 2025.10941393.
- [31] A. A. Jadhav and P. S. Pisal, "Harnessing Kernel Tricks for Nonlinear Problem Solving: SVM Applications," *2025 IRASET*, Fez, Morocco, 2025, pp. 1-7, doi: 10.1109/IRASET64571. 2025.11007960.
- [32] V. S. B. Chilluri et al., "Design and Implementation of a User Centered Design Framework for Software Development," in *Proc. 2025 CE2CT*, India, 2025, pp. 1232-1236, doi: 10.1109/CE2CT64011.2025.10941419.
- [33] B. P. Joshi et al., "Quaternion Intuitionistic Fuzzy Fusion Process: Applications to the Classification of Photo-Voltaic Solar Power Plants," *Int. J. Fuzzy Syst.*, vol. 27, pp. 713-733, 2025, doi: 10.1007/s40815-024-01798-w.